

External memory

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Magnetic Disk

- A magnetic disk is a storage device that uses a [magnetization](#) process to write, rewrite and access data. It is covered with a magnetic coating and stores data in the form of tracks, spots and sectors.
- Hard disks, zip disks and floppy disks are common examples of magnetic disks.

Magnetic Read and Write Mechanisms

- Data are recorded on and later retrieved from the disk via a conducting coil named the **head**; in many systems, there are two heads, a read head and a write head.
- The write mechanism exploits the fact that electricity flowing through a coil produces a magnetic field.
- Electric pulses are sent to the write head,

Data Organization and Formatting

- The head is a relatively small device capable of reading from or writing to a portion of the platter rotating under it.
- This gives rise to the organization of data on the platter in a concentric set of rings, called **tracks**.
- Each track is the same width as the head. There are thousands of tracks per surface.

Disk data layout

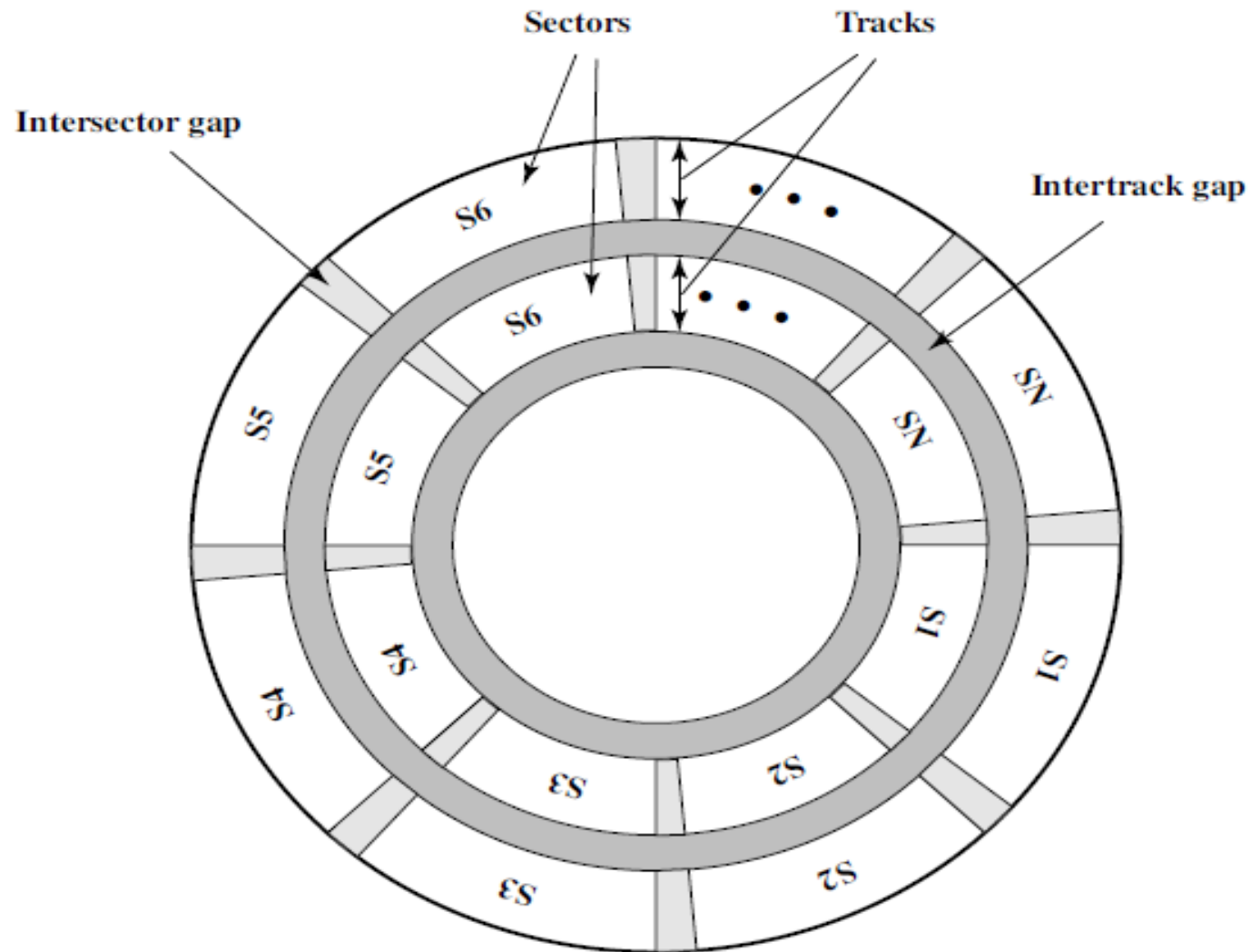


Figure 6.2 Disk Data Layout

Cont.

- Data are transferred to and from the disk in **sectors** (Figure 6.2). There are typically hundreds of sectors per track, and these may be of either fixed or variable length.
- In most modern systems, fixed-length sectors are used, with 512 bytes being the nearly universal sector size

Physical Characteristics

- Table 6.1 lists the major characteristics that differentiate among the various types of magnetic disks. First, the head may either be fixed or movable with respect to the radial direction of the platter. In a **fixed-head disk**, there is one read-write head per track. All of the heads are mounted on a rigid arm that extends across all tracks;
- such systems are rare today.

Cont.

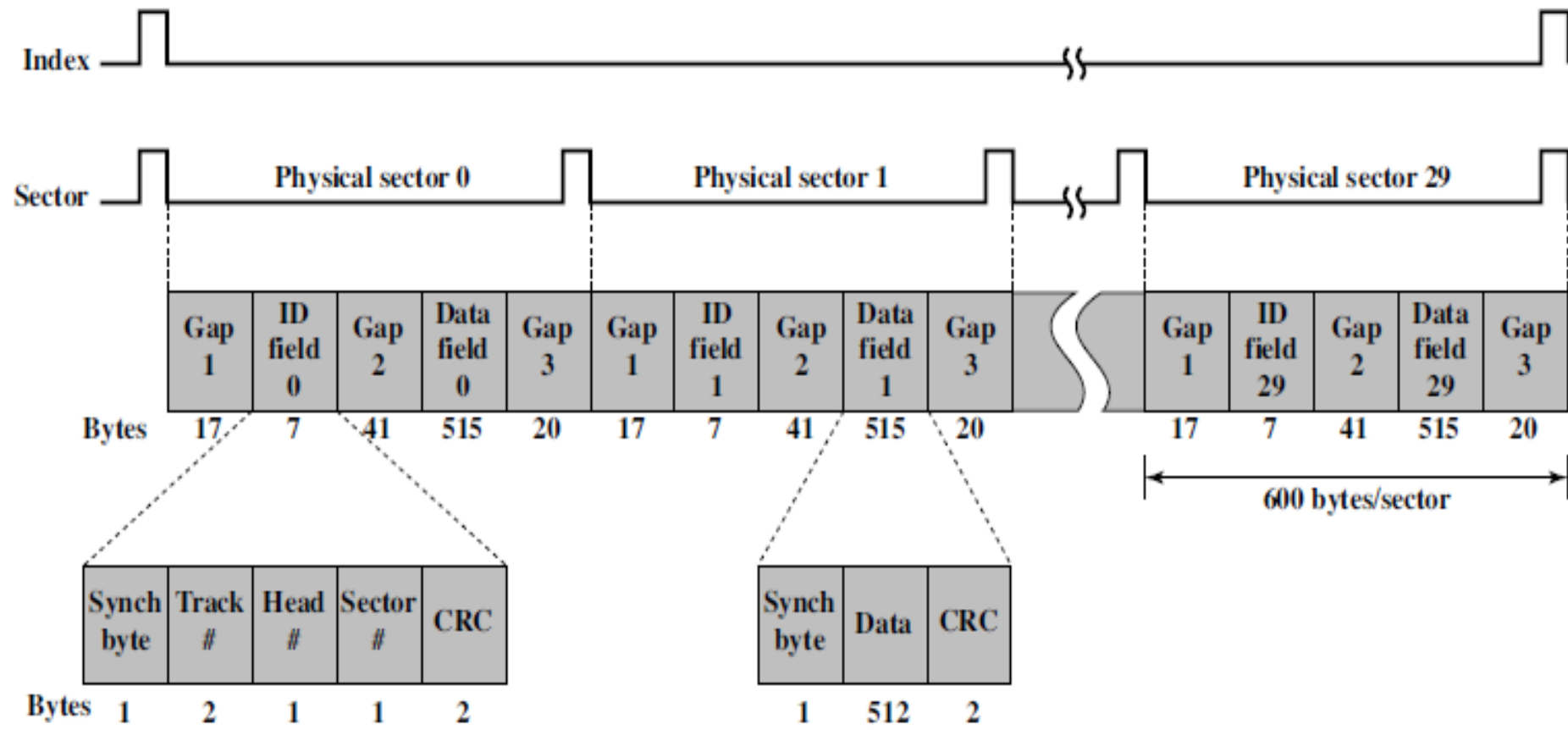


Figure 6.4 Winchester Disk Format (Seagate ST506)

Cont.

Table 6.1 Physical Characteristics of Disk Systems

Head Motion	Platters
Fixed head (one per track)	Single platter
Movable head (one per surface)	Multiple platter
Disk Portability	Head Mechanism
Nonremovable disk	Contact (floppy)
Removable disk	Fixed gap
	Aerodynamic gap (Winchester)
Sides	
Single sided	
Double sided	

Cont.

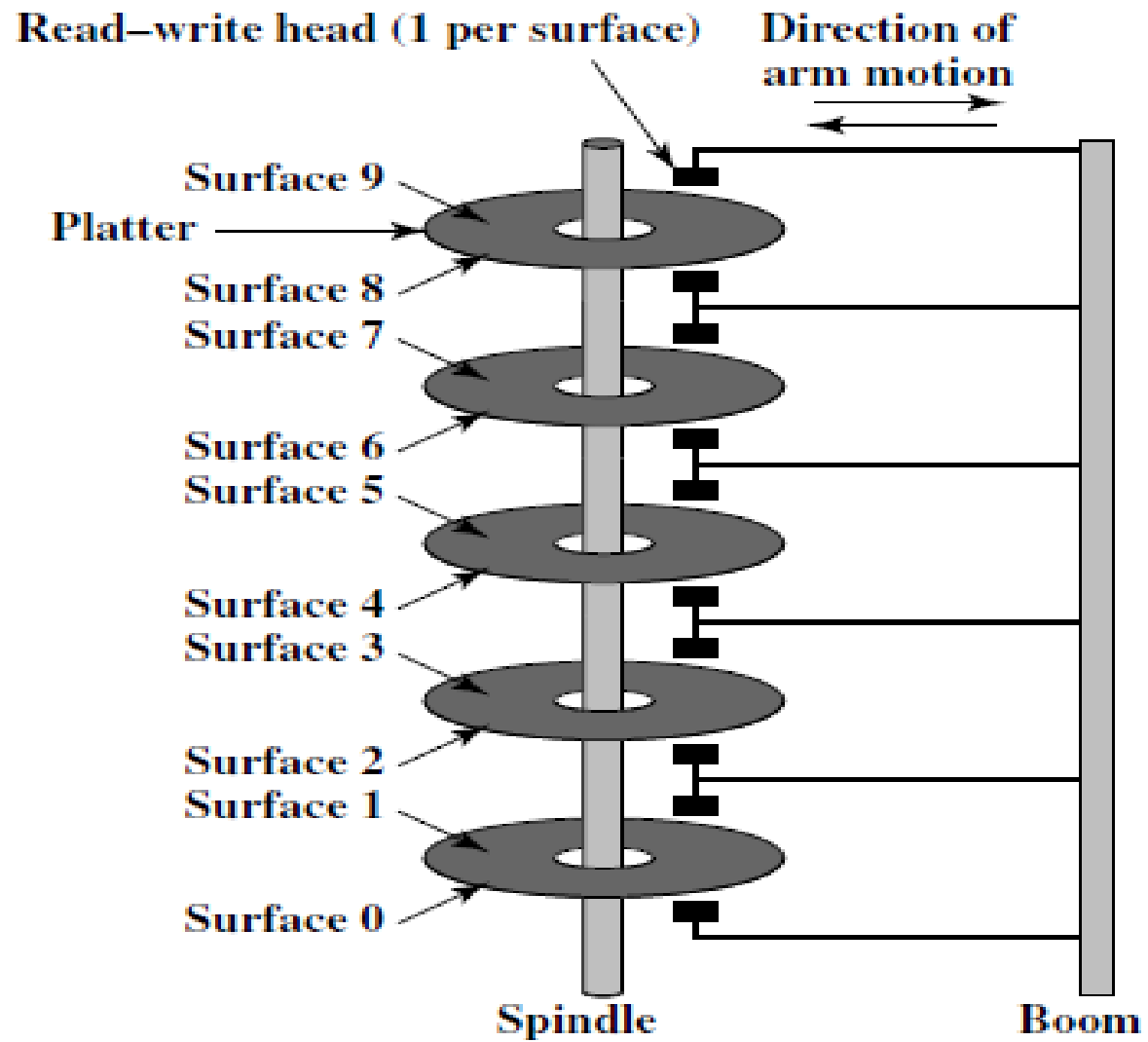


Figure 6.5 Components of a Disk Drive

Table 6.5 Optical Disk Products

CD

Compact Disk. A nonerasable disk that stores digitized audio information. The standard system uses 12-cm disks and can record more than 60 minutes of uninterrupted playing time.

CD-ROM

Compact Disk Read-Only Memory. A nonerasable disk used for storing computer data. The standard system uses 12-cm disks and can hold more than 650 Mbytes.

CD-R

CD Recordable. Similar to a CD-ROM. The user can write to the disk only once.

CD-RW

CD Rewritable. Similar to a CD-ROM. The user can erase and rewrite to the disk multiple times.

DVD

Digital Versatile Disk. A technology for producing digitized, compressed representation of video information, as well as large volumes of other digital data. Both 8 and 12 cm diameters are used, with a double-sided capacity of up to 17 Gbytes. The basic DVD is read-only (DVD-ROM).

DVD-R

DVD Recordable. Similar to a DVD-ROM. The user can write to the disk only once. Only one-sided disks can be used.

DVD-RW

DVD Rewritable. Similar to a DVD-ROM. The user can erase and rewrite to the disk multiple times. Only one-sided disks can be used.

Blu-Ray DVD

High definition video disk. Provides considerably greater data storage density than DVD, using a 405-nm (blue-violet) laser. A single layer on a single side can store 25 Gbytes.

RAID (Redundant Array of Independent Disks)

- To achieve greater performance and higher availability, servers and larger systems use RAID disk technology.
- RAID is a family of techniques for using multiple disks as a parallel array of data storage devices, with redundancy built in to compensate for disk failure.

Cont.

- The RAID scheme consists of seven levels, zero through six.
- These levels do not imply a hierarchical relationship but designate different design architectures that share three common characteristics:
 1. RAID is a set of physical disk drives viewed by the operating system as a single logical drive.
 2. Data are distributed across the physical drives of an array in a scheme known as striping, described subsequently.
 3. Redundant disk capacity is used to store parity information, which guarantees data recoverability in case of a disk failure.

Cont.

- The details of the second and third characteristics differ for the different RAID levels.
- RAID 0 and RAID 1 do not support the third characteristic.
- The term *RAID* was originally coined in a paper by a group of researchers at the University of California at Berkeley [PATT88].⁴

Table 6.3 RAID Levels

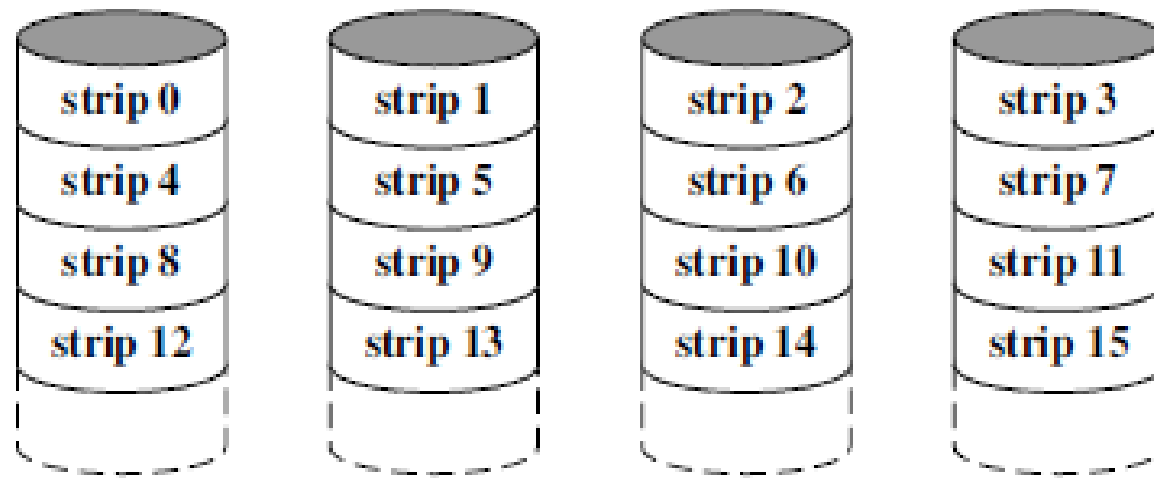
Category	Level	Description	Disks Required	Data Availability	Large I/O Data Transfer Capacity	Small I/O Request Rate
Striping	0	Nonredundant	N	Lower than single disk	Very high	Very high for both read and write
Mirroring	1	Mirrored	$2N$	Higher than RAID 2, 3, 4, or 5; lower than RAID 6	Higher than single disk for read; similar to single disk for write	Up to twice that of a single disk for read; similar to single disk for write
Parallel access	2	Redundant via Hamming code	$N + m$	Much higher than single disk; comparable to RAID 3, 4, or 5	Highest of all listed alternatives	Approximately twice that of a single disk
	3	Bit-interleaved parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 4, or 5	Highest of all listed alternatives	Approximately twice that of a single disk
Independent access	4	Block-interleaved parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 3, or 5	Similar to RAID 0 for read; significantly lower than single disk for write	Similar to RAID 0 for read; significantly lower than single disk for write
	5	Block-interleaved distributed parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 3, or 4	Similar to RAID 0 for read; lower than single disk for write	Similar to RAID 0 for read; generally lower than single disk for write
	6	Block-interleaved dual distributed parity	$N + 2$	Highest of all listed alternatives	Similar to RAID 0 for read; lower than RAID 5 for write	Similar to RAID 0 for read; significantly lower than RAID 5 for write

N = number of data disks; m proportional to $\log N$

RAID Level 0

- RAID level 0 is not a true member of the RAID family because it does not include redundancy to improve performance.
- However, there are a few applications, such as some supercomputers in which performance and capacity are primary concerns and low cost is more important than improved reliability.

Cont.



(a) RAID 0 (Nonredundant)

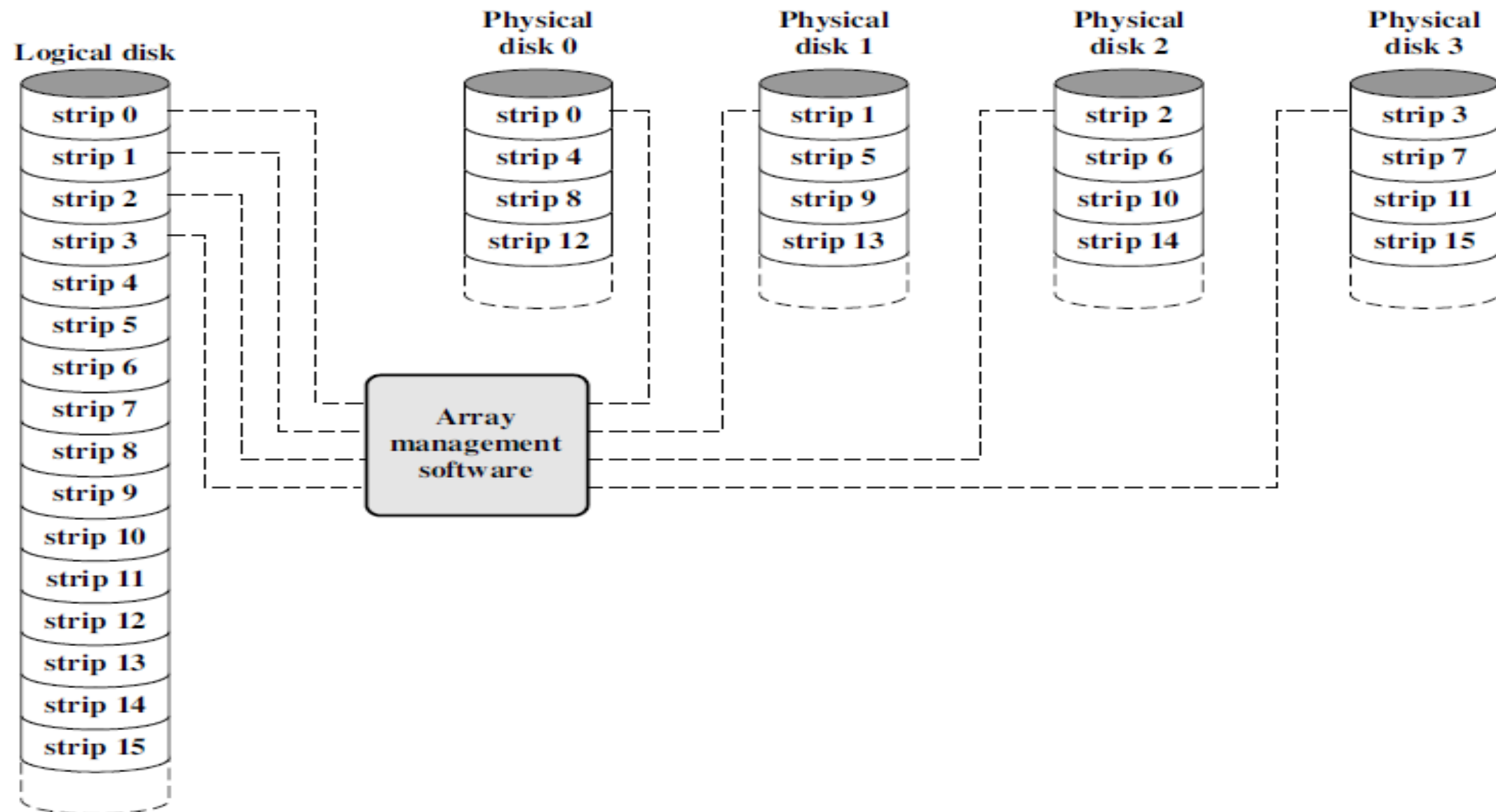
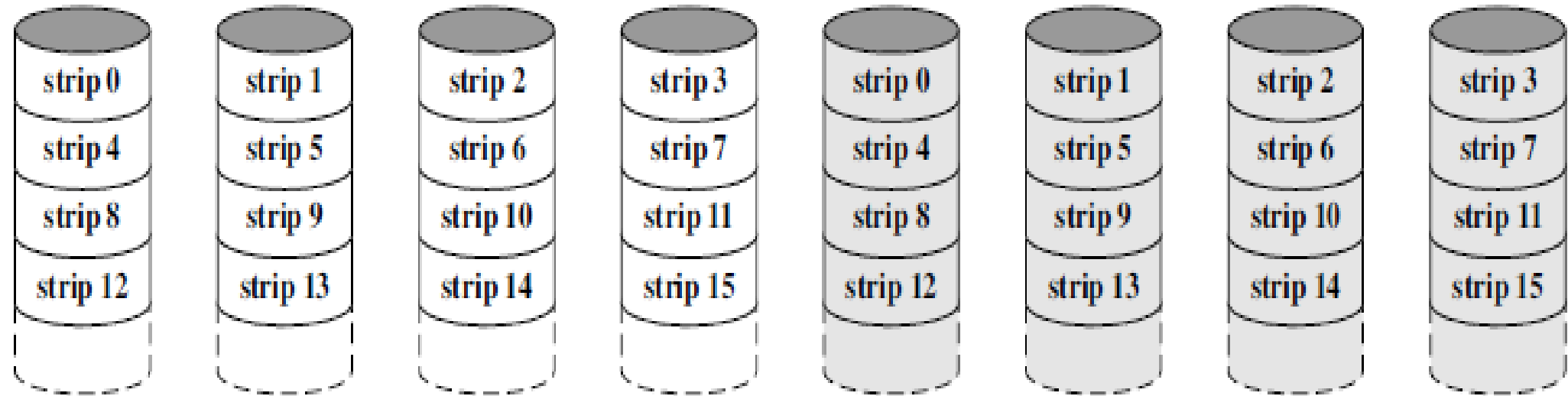


Figure 6.9 Data Mapping for a RAID Level 0 Array

RAID Level 1

- RAID 1 differs from RAID levels 2 through 6 in the way in which redundancy is achieved.
- In these other RAID schemes, some form of parity calculation is used to introduce redundancy, whereas in RAID 1, redundancy is achieved by the simple expedient of duplicating all the data.

Cont.



(b) RAID 1 (Mirrored)

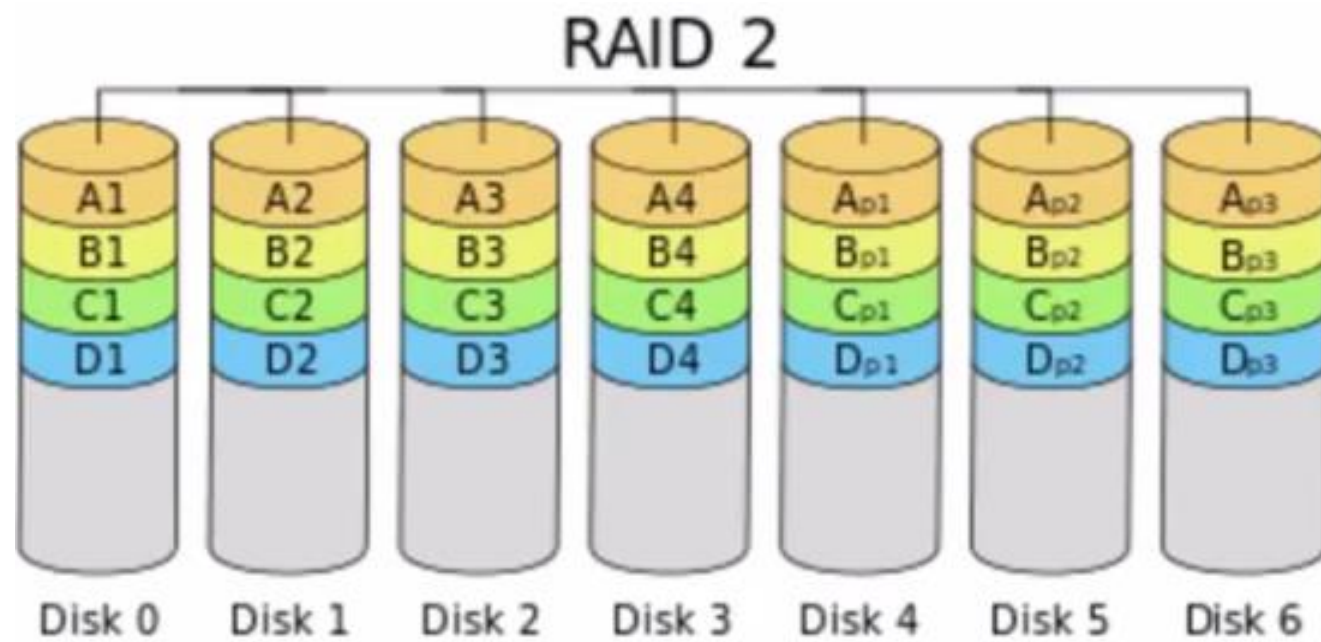
Advantage

1. A read request can be serviced by either of the two disks that contains the requested data, whichever one involves the minimum seek time plus rotational latency.
2. A write request requires that both corresponding strips be updated, but this can be done in
3. Recovery from a failure is simple. When a drive fails, the data may still be accessed from the second drive.

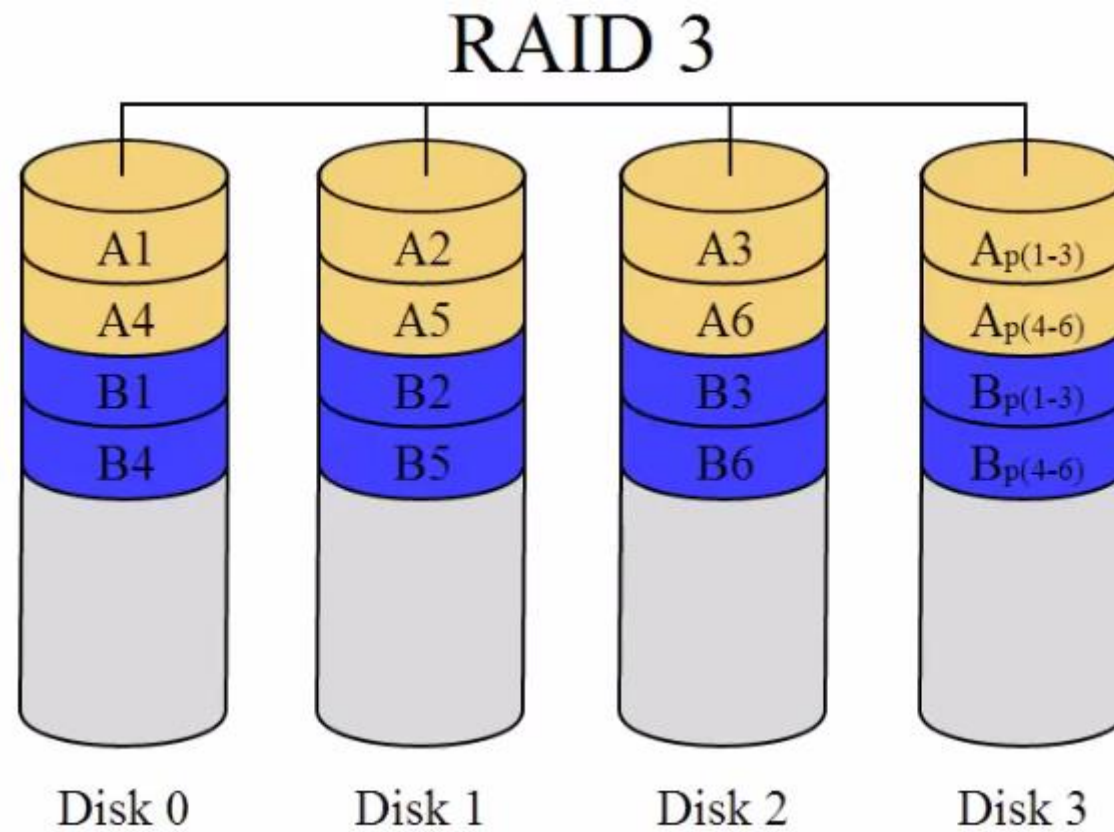
Dis advantage

- The principal disadvantage of RAID 1 is the cost; it requires twice the disk space of the logical disk that it supports.
- Because of that, a RAID 1 configuration is likely to be limited to drives that store system software and data and other highly critical files.

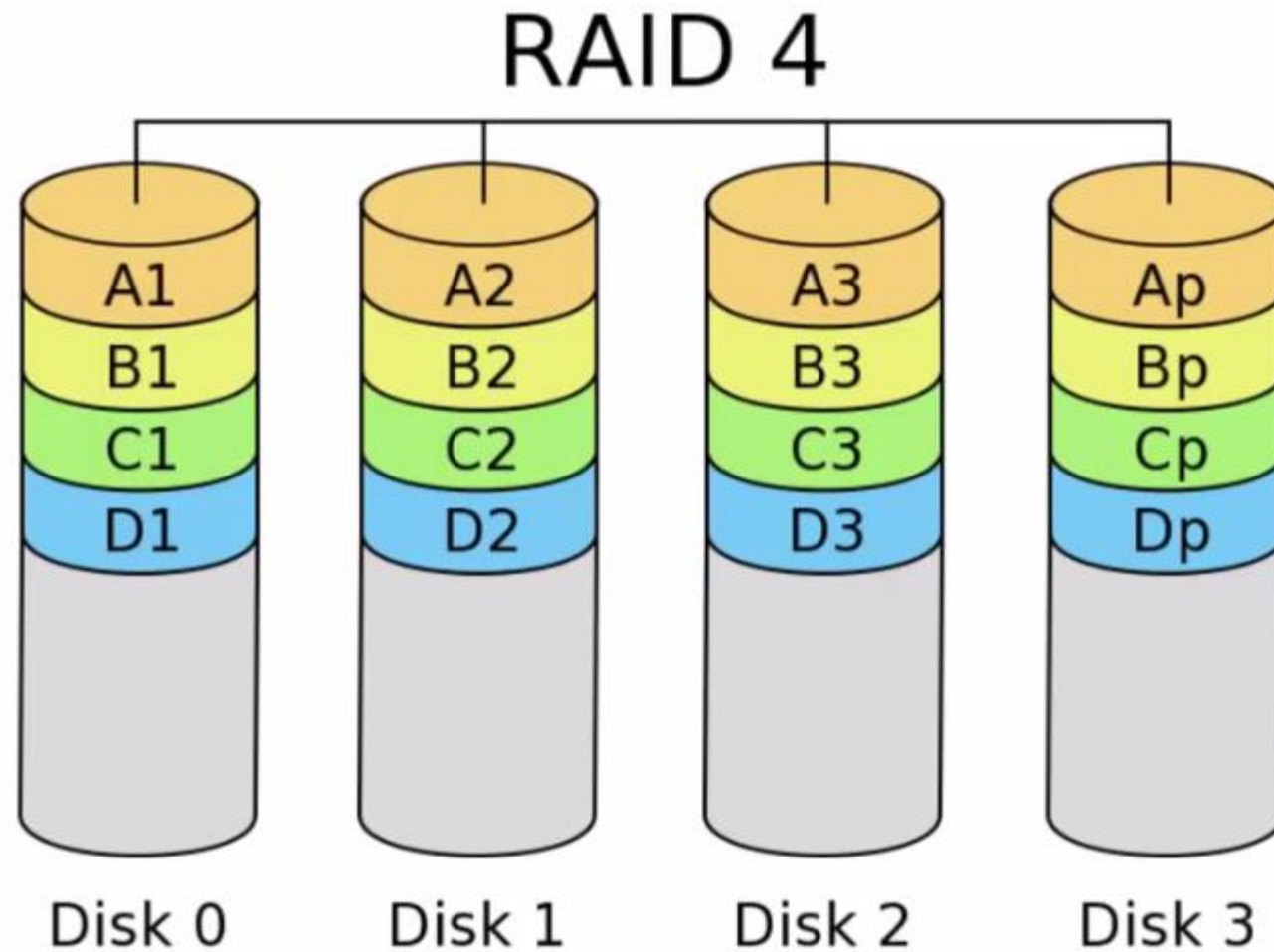
RAID-2



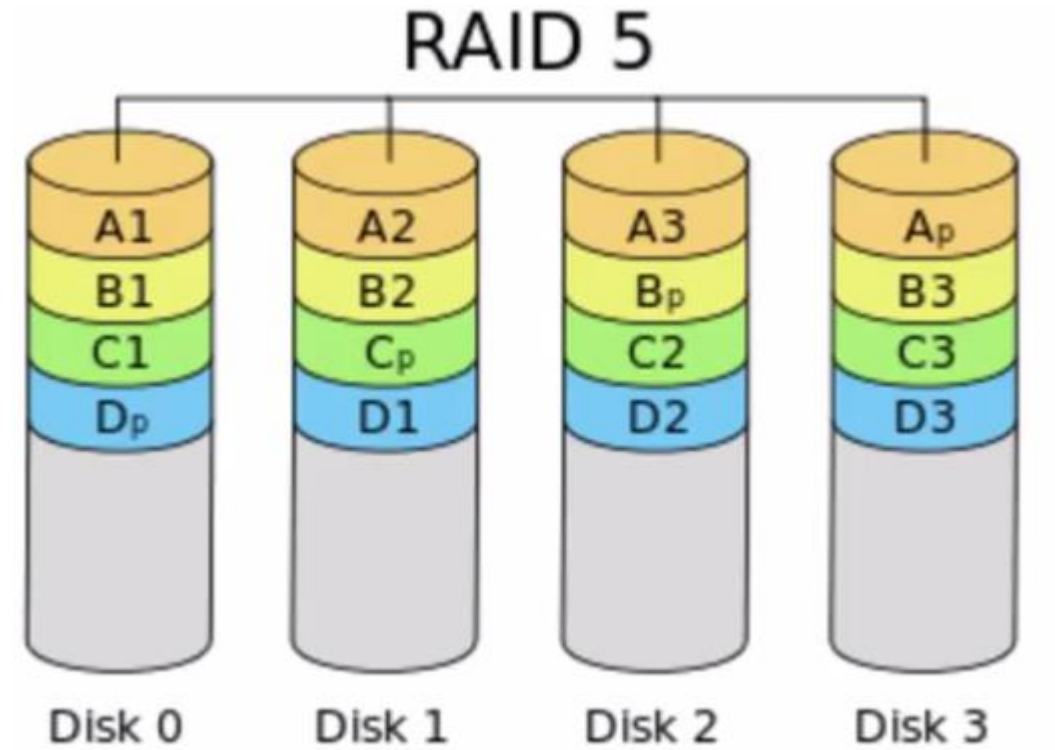
RAID-3



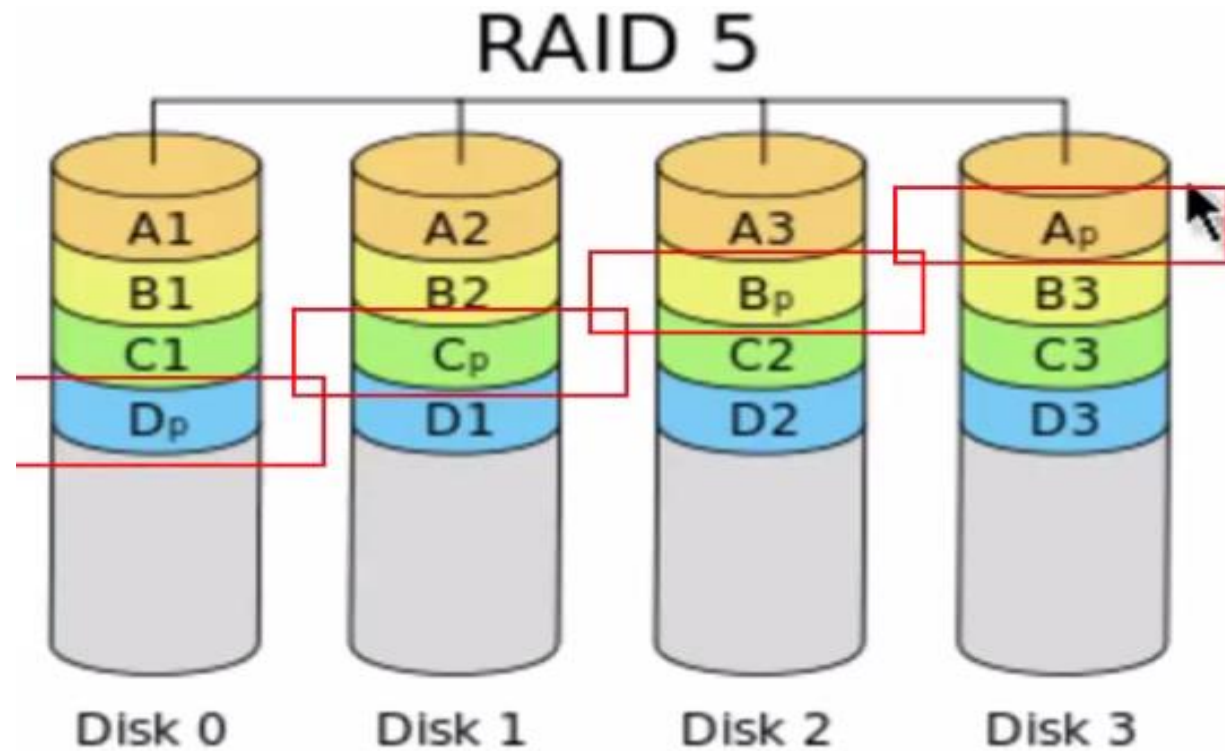
RAID-4



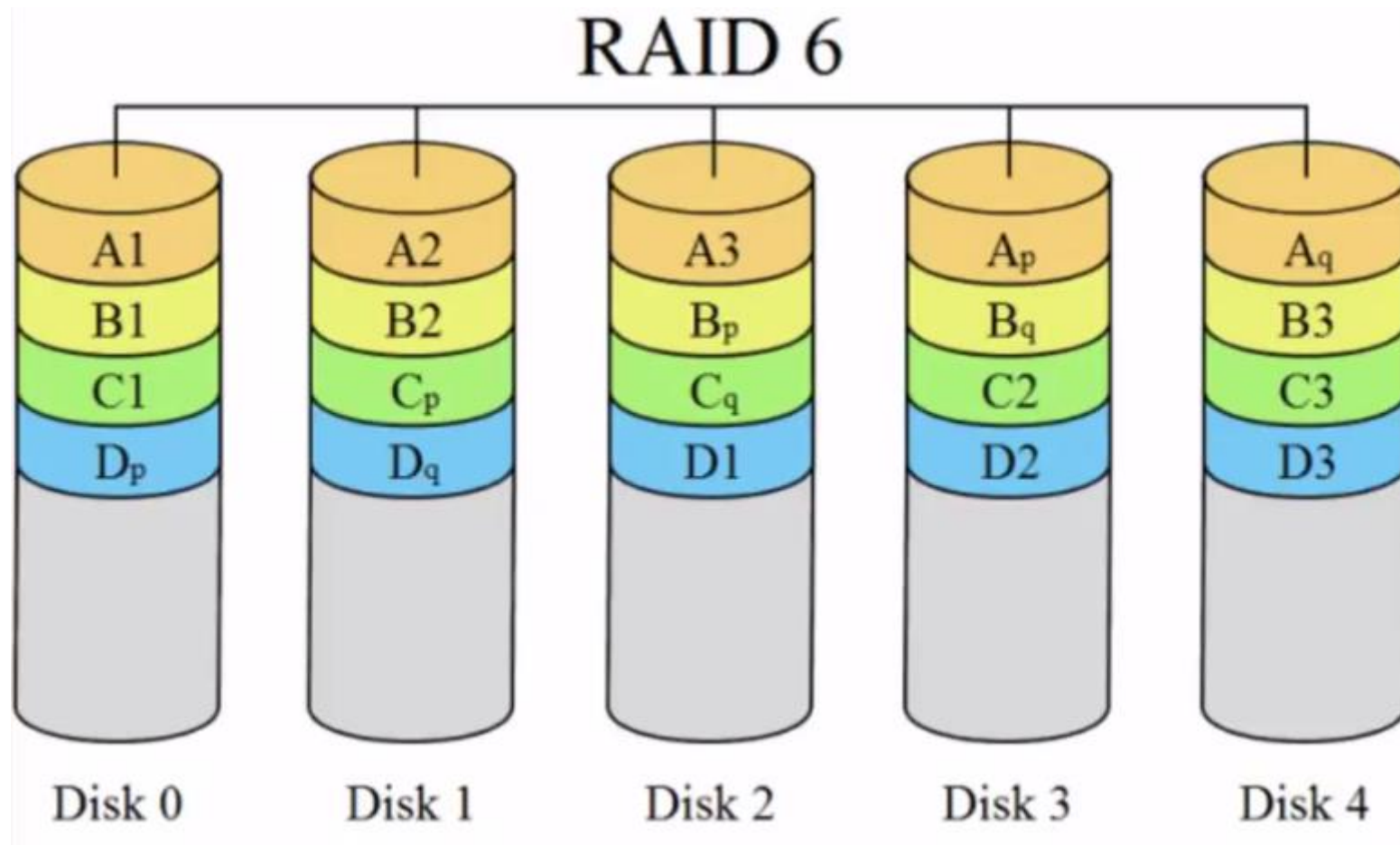
RAID-5



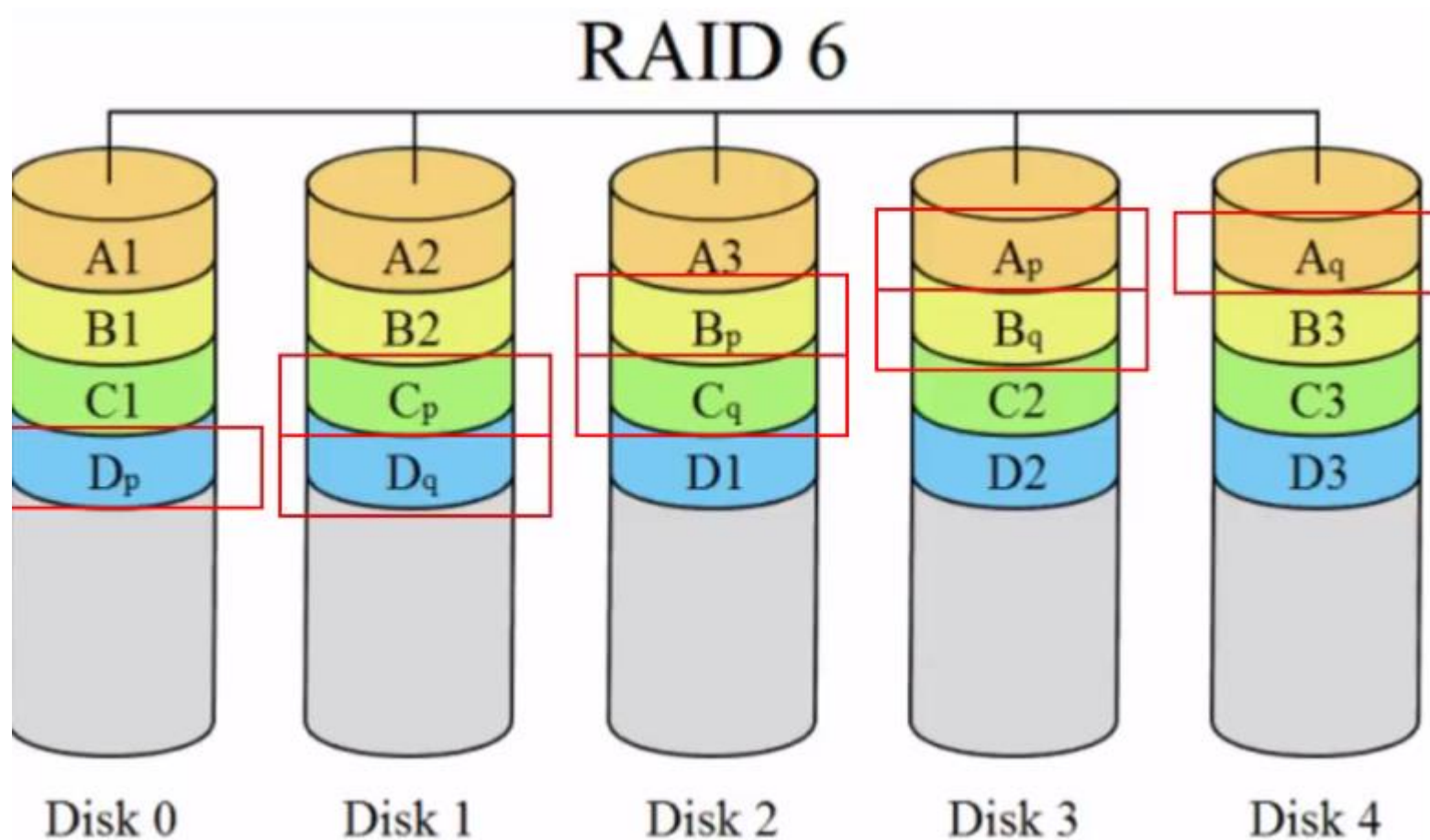
RAID-5 CONT.



RAID-6



RAID-6 CONT.



Thanks